Sandeep Singh Shekhawat

+91 8217399110 shekhawatsandeepsingh80@gmail.com linkedin.com/sandeep github.com/sandeep

Summary

Software Engineering professional at HCLTech with hands-on experience in Python, Django, backend development, REST APIs, automation, and DevOps. Experienced in developing and optimizing APIs, writing unit and API tests, supporting engineering workflows, and building internal automation solutions. Previously trained in Java Full Stack Development with Spring Boot, ReactJS, and PostgreSQL. Also experienced in AI/ML projects involving deep clustering, LLMs, and LangChain.

Education

Sri Sairam College of Engineering Bengaluru, India
B.E. in Artificial Intelligence and Machine Learning *2021-2025*

Sri Sairam Pre-University College Bengaluru, India
PCMCs - Physics, Chemistry, Mathematics, Computer Science *2019-2021*

Navodaya Educational Trust Bengaluru, India
2010-2019

Experience

HCLTech Jun,25 - Till now
Graduate Engineer Trainee / Software Engineering

- Completed a 4-month intensive training in **Full Stack Development**, gaining hands-on experience with Python, Java, Spring Boot, ReactJS, PostgreSQL, Git, and DevOps fundamentals.
- Contributed to a commercial web application project, applying backend, frontend, database, version control, and DevOps practices in a development environment.
- Transitioned into **Python/Django backend development**, working on API development, optimization, testing, and backend engineering tasks.
- Developed and optimized backend APIs with a focus on maintainability, reliability, performance, and clean implementation.
- Worked with **unit testing and API testing** to validate backend functionality and improve application reliability.
- Participated in planning and implementation of backend engineering tasks while collaborating with team members to deliver technical solutions.
- Currently working on **Python automation, DevOps practices, and internal engineering tools** to streamline development workflows and automate repetitive processes.

Projects

Advanced Market Segmentation using Deep Clustering | *Python, Deep Learning, Scikit-learn*

- Developed a customer segmentation solution using a **deep clustering approach** to identify meaningful customer groups from behavioral and spending-related data.
- Applied data preprocessing, feature scaling, representation learning, and clustering techniques to generate customer segments.
- Explored autoencoder-based dimensionality reduction and clustering to improve the quality of customer representations.

Chatbot for College Queries | *Python, LLM, LangChain*

- Developed an AI-powered chatbot designed to answer college-related queries using **LLM-based conversational processing**.
- Used **LangChain** to structure the application workflow and integrate language-model capabilities.
- Designed the solution to provide students with an interactive interface for retrieving information related to common college queries.

Robotic Arm Controlled using AI Chatbot | *Python, Arduino/ESP32, AI, IoT*

- Developed a prototype for remotely controlling a robotic arm through voice or text-based chatbot commands.
- Integrated Python-based command processing with microcontroller-controlled robotic hardware.
- Combined concepts from **AI, NLP, IoT, and robotics** to translate user commands into robotic arm operations.

Technical Skills

Languages: Python, Java, C++, SQL
Backend Development: Django, Django REST Framework, Flask, REST APIs, Backend Development
Testing: Unit Testing, API Testing, pytest, Postman, Bruno
Databases: PostgreSQL, SQL
AI / Machine Learning: Machine Learning, Deep Learning, NLP, LLMs, LangChain, OpenCV
Frontend: ReactJS, HTML, CSS
Automation & DevOps: Python Automation, DevOps, DevOps Fundamentals
Frameworks: Spring, Spring Boot
Version Control: Git, GitHub

Awards

Py Hunt Winner, *Webathon*

- Secured first prize in an inter-college coding competition comprising two rounds: the first round focused on dry run coding, while the second round involved building a web application.

Best Novelty Award in 24hr Hackathon

- Developed a home automation solution by embedding IoT applications with AI, which helped receive the Best Novelty Award.